Project Design Phase
Proposed Solution Template

Date	15 February 2025
Team ID	GACSE-109
Project Name	AI Based Exam Anxiety Detector
Maximum Marks	2 Marks

Proposed Solution Template

S.No.	Parameter	Description
	1. **Problem Statement** (Problem to be solved)	Students often experience significant anxiety before and during examinations due to academic pressure, competition and fear of failure. However, many students struggle to openly express their stress or seek help, making it difficult for educators and institutions to identify those who need support. Traditional approaches such as surveys or manual observation are often slow, subjective, and unable to detect anxiety in real-time. Therefore, there is a need for an intelligent system that can automatically identify exam-related anxiety early and help students manage their mental well-being.
	2. **Idea / Solution description**	The proposed solution is an **AI-Based Exam Anxiety Detector**, an intelligent mental-wellness support system that analyzes student-generated text such as reflections, feedback, or pre-exam thoughts to detect exam-related anxiety. The system uses Natural Language Processing (NLP) 4 a **BERT-based deep learning model** to analyze linguistic patterns associated with stress, nervousness, fear, and emotional pressure. Based on the analysis, the system classifies anxiety into three levels. Low Anxiety ▫ Moderate Anxiety ▫ High Anxiety
	3. **Novelty / Uniqueness**	• Uses AI and **deep learning (BERT)** for automatic detection of exam anxiety. • Provides real-time analysis instead of traditional surveys. • Detects emotional **stress directly from** student text input. • Offers **instant feedback** with anxiety-management tips: • Allows educational institutions to monitor anonymized anxiety trends.
	4. **Social Impact / Customer Satisfaction**	• Helps students **identify** and manage exam anxiety early. • Encourages **mental health** awareness in **educational environments**. • Supports institutions in providing **timely counseling and academic support**. • Improves student **well-being**, confidence, and **academic performance**. • Creates a supportive and stress-aware **educational ecosystem** such as extrends and patterns.
	5. **Business Model** (Revenue Model)	The solution is highly scalable due to its AI-driven and cloud-based architecture. • Can be integrated with **Learning Management Systems (LMS)** used by universities. • Can support thousands of students simultaneously through **cloud deployment**. • The NLP model can be trained with more **datasets to improve accuracy**.